3D CT Image Visualization using Blender

Forschungspraktikum in Medical Engineering

submitted
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Übersicht

Dieses Projekt präsentiert einen wiederverwendbaren Workflow zur Visualisierung von CT-Daten mit Blender und Bioxel Nodes. Dafür wurde ein node-basiertes Template entwickelt, das Rendering und Farbzuweisung für volumetrische medizinische Daten automatisiert. Durch die Anpassung von Visualisierungsparametern können unterschiedliche anatomische Strukturen hervorgehoben werden, ohne die zugrunde liegende Pipeline zu verändern. Die Ergebnisse zeigen, dass Blender eine flexible und effiziente Umgebung für die wissenschaftliche Visualisierung von CT-Datensätzen bieten kann.

Abstract

This project presents a reusable workflow for CT image visualization using Blender and Bioxel Nodes. A node-based template was developed to automate rendering and color mapping for volumetric medical data. By adjusting visualization parameters, different anatomical structures can be emphasized without modifying the underlying pipeline. The results demonstrate that Blender can provide a flexible and efficient environment for scientific visualization of CT datasets.

Contents

Chapter 1

Introduction

Medical imaging plays a central role in modern healthcare and biomedical research. Among various imaging modalities, Computed Tomography (CT) is widely used due to its ability to provide high-resolution volumetric representations of internal anatomical structures. Although CT data is typically analyzed as a sequence of two-dimensional slices, three-dimensional visualization offers a more intuitive understanding of spatial relationships and structural complexity.

With the increasing demand for effective visualization techniques, modern computer graphics tools have been explored for scientific applications. Blender, an open-source 3D graphics platform, provides advanced rendering capabilities and supports procedural workflows that enable flexible scene construction. These features make Blender a promising environment for volumetric data visualization.

Recent developments in Blender extensions have further expanded its applicability to biomedical data. In particular, the Bioxel Nodes add-on introduces a node-based framework for processing and rendering volumetric datasets. This approach allows users to construct visualization pipelines by connecting modular processing steps, enabling both flexibility and reusability.

However, creating effective visualizations still requires manual configuration of parameters such as thresholding, color mapping, and rendering settings. This process can be time-consuming and may lead to inconsistent results across different datasets. Therefore, there is a need for a more efficient and standardized workflow.

In this project, a reusable visualization pipeline is developed in Blender using Bioxel Nodes. The pipeline is implemented as a template that integrates data processing, color mapping, and rendering into a unified node structure. Once the template is configured, users only need to import a CT dataset, after which all processing steps are automatically applied. This significantly reduces manual effort and ensures consistent visualization quality.

The main contribution of this work is the design of a template-based workflow that separates pipeline configuration from data input. This separation enables efficient reuse of visualization settings and facilitates rapid generation of high-quality 3D renderings. The proposed approach demonstrates how node-based workflows in Blender can streamline volumetric data visualization and support practical applications in scientific communication and education.

Chapter 2

Background

2.1 CT Image Data

Computed Tomography (CT) is a medical imaging technique that reconstructs cross-sectional images of the human body using X-ray measurements acquired from multiple angles. A CT scan produces a stack of two-dimensional slices that together form a volumetric dataset.

Each element in the volume, known as a voxel, represents a small three-dimensional unit with an associated intensity value corresponding to tissue density. These values are commonly expressed in Hounsfield Units (HU), which allow different tissue types such as bone, soft tissue, and air to be distinguished.

The volumetric nature of CT data enables advanced visualization techniques beyond slice-based inspection. By analyzing voxel intensities, relevant anatomical structures can be highlighted, making CT particularly suitable for volume rendering approaches.

2.2 Blender for Scientific Visualization

Blender is an open-source 3D graphics software widely used for modeling, rendering, and animation. In recent years, it has also been adopted for scientific visualization due to its powerful rendering engines and extensible architecture.

One of Blender's key advantages is its support for procedural workflows, allowing complex scenes to be constructed through reusable node-based systems. This makes it particularly suitable for handling structured data such as volumetric datasets.

Through its extensibility and support for add-ons, Blender can be adapted for specialized tasks, including the visualization of medical imaging data.

2.3 Bioxel Nodes

Bioxel Nodes is a Blender add-on for processing volumetric biomedical data directly inside Blender. It provides a node-based workflow for importing, manipulating, and rendering voxel datasets.

Its node structure allows users to build visualization pipelines from components such as data import, filtering, thresholding, and color mapping. This modular design makes it easy to adjust parameters and test different visualization strategies.

Bioxel Nodes also supports volume rendering by mapping voxel intensity values to color and opacity. This enables relevant anatomical structures to be emphasized while less important regions are visually suppressed.

Together with Blender's rendering capabilities, Bioxel Nodes provides a practical framework for creating customizable and reusable medical visualization workflows.

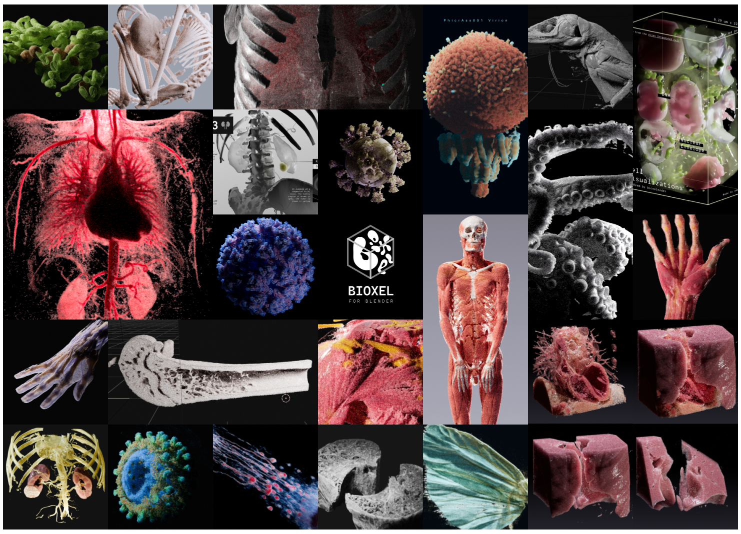


Figure 2.1: Examples of volumetric rendering using Bioxel Nodes (source: official Bioxel Nodes documentation [Omo25]).

Chapter 3

Methodology

3.1 Design Goals

The goal of this project is to develop a reusable and efficient workflow for visualizing CT volumetric data in Blender. Instead of manually configuring visualization parameters for each dataset, the proposed approach focuses on designing a template-based pipeline that can be applied consistently across different inputs.

The workflow aims to reduce manual effort, ensure consistent visual quality, and enable rapid generation of results. By separating pipeline configuration from data input, the system allows users to reuse predefined settings and focus on data exploration rather than repeated setup.

3.2 Pipeline Overview

The visualization workflow consists of four main stages: data import, volumetric processing, transfer function design, and rendering.

The pipeline is implemented within a node-based framework using Bioxel Nodes. CT data is first imported as a volumetric dataset, followed by preprocessing operations to enhance relevant structures. A transfer function is then applied to map voxel intensity values to visual properties such as color and opacity. Finally, rendering settings are used to generate the output image.

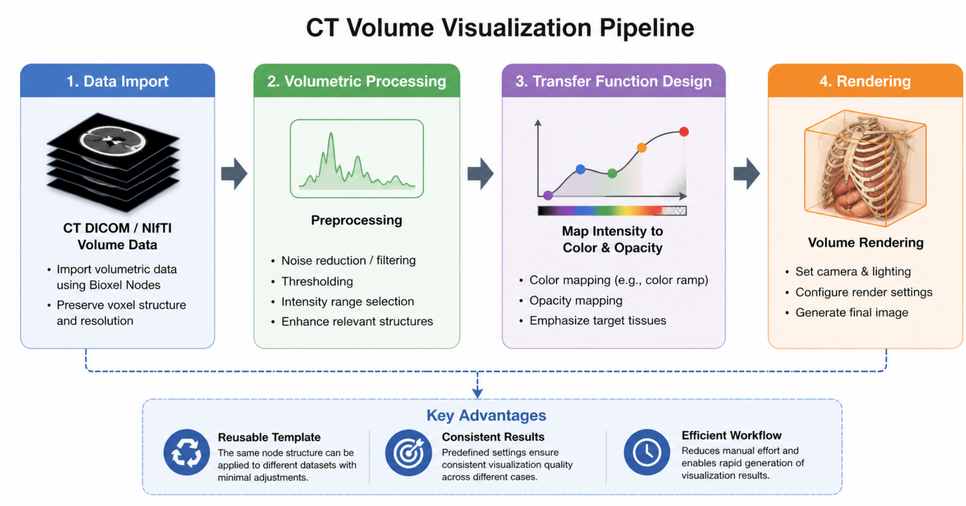


Figure 3.1: Overview of the visualization pipeline. The workflow includes data import, preprocessing, transfer function mapping, and rendering.

3.3 Data Import and Volume Setup

CT datasets are imported into Blender using Bioxel Nodes as volumetric data. During this process, voxel structure and spatial resolution are preserved to maintain anatomical accuracy.

Once the data is loaded, it is automatically connected to the predefined node structure of the template. This ensures that all subsequent processing steps are applied consistently without requiring manual configuration.

3.4 Template-based Processing and Color Mapping

The core of the workflow is the template implemented using Bioxel Nodes. The node graph is organized into functional components such as filtering, thresholding, and color mapping.

Preprocessing steps are applied to reduce noise and enhance relevant structures. Thresholding is used to isolate anatomical regions based on voxel intensity values. A transfer function is then defined to map these values to color and opacity, enabling visual differentiation of tissue types.

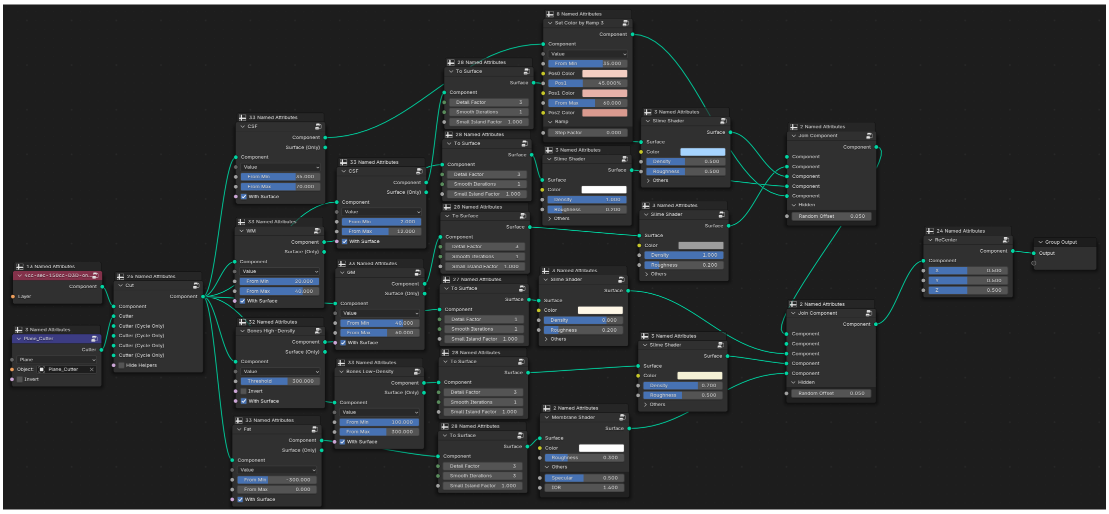


Figure 3.2: Node-based template implemented in Bioxel Nodes. The graph is structured into modular components for processing and visualization.

3.5 Multiple Visualization Configurations

Two visualization configurations were developed to demonstrate the flexibility of the template.

The first configuration focuses on soft tissue visualization. By selecting appropriate intensity ranges and applying smooth opacity transitions, this setup highlights muscles and surrounding anatomical structures.

The second configuration is designed for brain tissue visualization. In this setup, voxel intensity ranges are used to distinguish between white matter, gray matter, and cerebrospinal fluid (CSF). Color mapping is applied to emphasize structural differences.

Both configurations are implemented using the same template, requiring only parameter adjustments.

3.6 Rendering Configuration

Rendering parameters such as camera position, lighting, and rendering engine settings are predefined in the template. This ensures consistent output quality across different datasets.

The final images are generated directly in Blender and can be exported for further use.

Chapter 4

Results

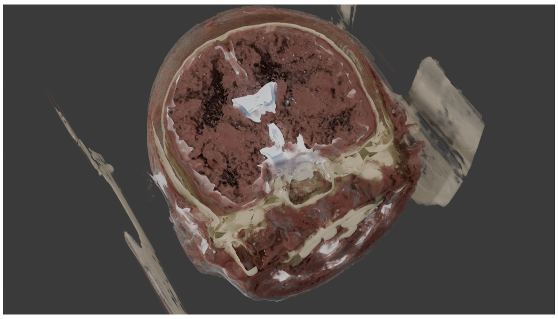
4.1 Visualization Results

The proposed pipeline was evaluated using two different visualization configurations. Figure 4.1 presents a side-by-side comparison of the generated results, illustrating how parameter adjustments influence the appearance of volumetric CT data.

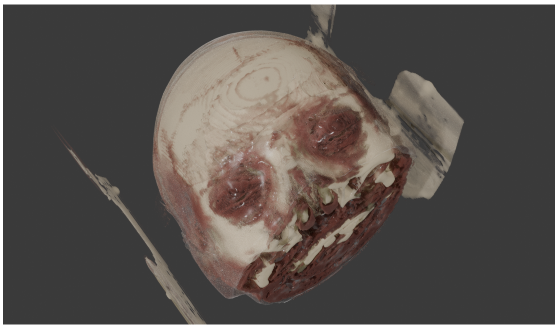
The top row (Figure 4.1a and b) presents the soft tissue-focused configuration. In this setup, smooth opacity transitions and continuous intensity mappings are used to emphasize outer anatomical structures such as muscle and soft tissue layers. As a result, the visualization appears more coherent and preserves the overall shape of the head, making it easier to interpret global anatomical features.

In contrast, the bottom row (Figure 4.1c and d) shows the brain-focused configuration. Here, intensity thresholds are adjusted to separate different tissue regions, resulting in a more segmented visualization. Internal structures become more distinguishable, and variations in tissue density are more clearly represented.

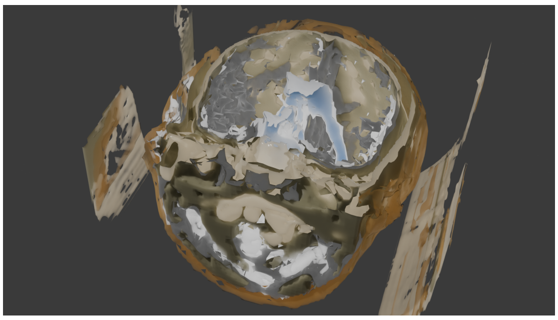
Comparing the two configurations, it can be observed that the soft tissue approach prioritizes visual continuity and surface perception, while the brain-focused approach emphasizes structural differentiation and internal detail. These differences are achieved by modifying parameters such as threshold ranges, color mapping, and opacity, without altering the underlying node-based pipeline.



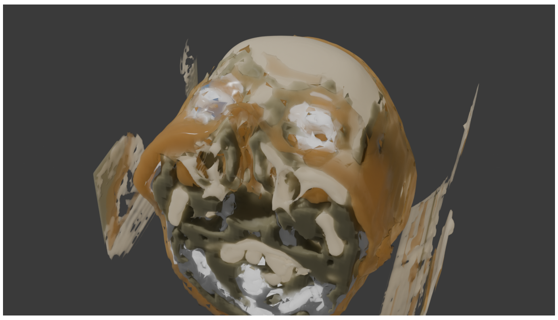
(a) Soft tissue visualization A



(b) Soft tissue visualization B



(c) Brain visualization A



(d) Brain visualization B

Figure 4.1: Comparison of visualization results using two configurations. The top row shows soft tissue-focused rendering, while the bottom row demonstrates brain structure visualization with enhanced tissue separation.

4.2 Template Efficiency

A key advantage of the proposed workflow is efficiency. Once the template is configured, the user only needs to import a CT dataset to generate a visualization.

Compared to traditional workflows that require repeated parameter tuning, the template significantly reduces manual effort. In addition, the use of predefined rendering settings ensures consistent output quality across different datasets.

Another benefit of the approach is reproducibility. Since the visualization process is defined by a fixed node structure, the same configuration can be applied to multiple datasets with minimal adjustments. This makes the pipeline particularly useful for comparative studies and large-scale data processing.

Overall, the results confirm that the proposed pipeline provides an efficient, flexible, and reusable solution for CT data visualization in Blender.

Chapter 5

Discussion

The results demonstrate that Blender together with Bioxel Nodes can be used as an effective platform for volumetric CT visualization. By constructing a reusable node-based template, different visualization styles can be generated from the same dataset through parameter adjustment alone. This highlights the flexibility of the proposed workflow and shows how node-based systems can simplify the visualization process for medical imaging data.

One important observation is the strong influence of parameter selection on the final rendering result. Small changes in intensity thresholds, opacity, or color mapping can significantly alter the appearance of anatomical structures. In the soft tissue-focused configuration, smooth opacity transitions create a coherent and visually continuous representation of the head. In contrast, the brain-focused configuration emphasizes internal structures by separating tissue regions according to voxel intensity. These differences demonstrate how the same underlying data can be interpreted in multiple ways depending on the visualization objective.

Another advantage of the proposed approach is reusability. Once the pipeline structure is prepared, the user only needs to import a new CT dataset to generate visualizations with minimal manual modification. Compared to workflows that require repeated setup and parameter tuning, the template-based approach improves efficiency and produces more consistent results across datasets.

The project also shows the potential of Blender as a scientific visualization environment. Unlike traditional medical imaging software that mainly focuses on diagnostic viewing, Blender provides advanced rendering tools, flexible material systems, and procedural workflows that allow visually expressive representations of volumetric data. This makes it particularly suitable for educational visualization, scientific communication, and presentation purposes.

Despite these advantages, several limitations remain. The visualization quality still depends heavily on manual parameter adjustment, especially when datasets contain noise or inconsistent intensity distributions. In some cases, artifacts and unwanted structures may appear due to imperfect threshold selection or limitations in the voxel data itself. Furthermore, the workflow focuses primarily on visualization and does not include automated segmentation or quantitative medical analysis.

Future work could improve the pipeline by integrating automatic tissue segmentation or adaptive threshold selection methods. Additional improvements may also include interactive visualization controls, animation support, and optimization for larger datasets. Combining Blender-based rendering with machine learning approaches could further enhance the automation and consistency of medical image visualization workflows.

Overall, the project demonstrates that reusable node-based pipelines in Blender provide a flexible and efficient solution for CT image visualization. The combination of procedural workflows and high-quality rendering capabilities enables the creation of customizable visual representations that can support scientific visualization and medical communication.

List of Figures

Bibliography

[Omo25] OmooLab. Bioxel nodes documentation. https://omoolab.github.io/BioxelNodes/1.0.x/, 2025. Accessed: 2026-05-17 (cited on p. 4).